The TikTok Effect on Films:

Similarity and Topic Diversity in Movie Subtitles Over Time

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ABSTRACT

The rise of short-form video platforms, most notably TikTok's mainstream adoption in Western markets around 2018, has been widely associated with a structural shift in audience attention patterns. A growing body of research suggests that habitual consumption of short-form content is linked to reduced sustained attention and a preference for faster, more repetitive narrative stimuli (Lin et al., 2024). If this behavioural shift has permeated mainstream cinema, we would expect to observe measurable changes in the narrative structure of popular films over time — specifically, an increase in dialogue redundancy and thematic concentration as filmmakers adapt to audiences less tolerant of sustained narrative complexity.

This study investigates whether such discontinuity is detectable in the textual content of movie scripts. Using sentence embeddings generated by the all-MiniLM-L6-v2 SBERT model and cosine similarity as the primary analytical tool, we construct a Topic Diversity Index (TDI) for 746 films released between 2003 and 2025. Our central hypothesis is that films released after 2018 will exhibit greater narrative repetitiveness (similar sentences) and thematic concentration (fewer topics) than those released before, reflecting the influence of shortened audience attention spans on mainstream screenwriting conventions. We treat 2018 as the pre/post comparison threshold, given TikTok, Instagram, Facebook, and YouTube's incorporation of short-form video into their platforms and their impact on media consumption, marking a plausible structural break in content pacing expectations.

RELATED WORK

SBERT was the primary model used to produce sentence embeddings, and cosine similarity was applied to compute sentence similarities, together generating each movie subtitle's Cosine Similarity and Topic Diversity Index. SBERT is foundationally built on BERT – Bidirectional Encoder Representations from Transformers (Devlin et al., 2019). It is a Natural Language Processing (NLP) model that looks at the left and right side of a word simultaneously to learn context thoroughly and is trained by predicting randomly masked words according to its surrounding terms (Devlin et al., 2019). SBERT is a further development of BERT that handles sentences rather than just words using sentence embeddings. It comprises a siamese network architecture – two identical BERT models trained on sentence pairs simultaneously – that allows sentences with semantically similar meanings to be assigned numerically closer vectors. This allows for efficient vector comparisons using cosine similarity (Reimers & Gurevych, 2019).

A study by Sharma et al. used this tandem to analyse query similarities in the Amazon Q&A Dataset, providing users with appropriate answers regardless of phrasing or grammatical variation. They particularly used the all-MiniLM-L6-v2 SBERT model, the same model used in our study. Sharma et al. concluded that semantic systems produce more accurate outputs and answers over keyword-specific search systems (Sharma et al., 2024). A similar approach was used to build semantic search engines for movies, enabling users to receive semantically similar recommendations by inputting their desired movie plot (Mustafa, 2024).

The all-MiniLM-L6-v2 model was similarly used to collect and rank relevant literature in response to a user query comprising a title and an abstract prompt. It was concluded that the MiniLM model performed best of the three models due to its 'general-purpose' use, whereas the other models had specialised uses (e.g., for scientific papers) (Dhakal et al., 2025). Studies consistently show that SBERT and cosine similarity are accurate and efficient models for identifying semantically similar sentences across diverse domains such as movie scripts and academia; an important feature when user queries with similar objectives are phrased differently. However, existing applications primarily revolve around user query recommendations. Our study takes a new approach by applying SBERT and cosine similarity to generate a Topic Diversity Score for movie scripts, bridging a methodological gap in the literature.

METHODOLOGY

Data Collection and Preprocessing (NB00_extraction_subtitles)

The primary source used for data collection was the "OpenSubtitles" website, which contains a subtitle text corpus for a vast range of movies. Specifically, subtitles were collected by querying a subtitles endpoint from the OpenSubtitles REST API, filtered by language (English) and sorted by download count (descending). We collected the movie subtitles with the highest download count from 2003 to 2025 (25-50 films per year), with the result being 787 raw SubRip Subtitle (SRT) files across this time range. The corpus was subsequently enriched with metadata from The Movie Database (TMDb) API for each of these films. The SRT files had an IMDb ID, which was used as a lookup key. Thus, each movie was matched to its genre, runtime, budget, ratings, and the number of ratings. We determined 2018 as our pre/post comparison cutoff given the mainstream incorporation of TikTok into Western media consumption, marking a structural shift in audience attention patterns and content pacing expectations (The Guardian, 2022).

For preprocessing, the parse_srt() function was used to clean the previously downloaded raw SRT files. After identifying common issues, a preprocessing pipeline was implemented to strip timestamp lines, subtitle index numbers, HTML tags, sound descriptions (in brackets), speaker labels, music labels, and leading dialogue dashes. HTML entities were also decoded (for example, & converted to &).

All the remaining lines were then concatenated into a single, clean string for each movie. The individual dialogue lines for each film were retained, and some basic details, such as word, line, and character counts, were also computed. The output of this entire process was then saved as a CSV file titled "clean_texts.csv" serving as a starting point for our analysis.

Sentence Embeddings Using SBERT (NB01_sbert_embedding)

Before applying SBERT to our sentence-level corpus, several assumptions about the dataset were made. Scripts contain stage directions along with dialogue; thus, we had to specify row value limits as quotation marks to only filter in dialogues. Assuming that one quotation (row entry) may consist of several sentences, a function was created to split row values according to punctuation marks such as "!", ".", and "?". Assumptions were also made that certain words like titles/honorifics (e.g. "Mr.", "Mrs Maria", etc.) or single-word responses (e.g. "No!", "Ok!") would be present in the dialogue, which the function may incorrectly code as full sentences. To ensure that sentences were meaningful, a sentence-length limit of more than 3 words (>3) was set to prevent the model from focusing on or producing an abundance of less meaningful sentences. Finally, assuming the values were still a list/array of sentences, the explode() function was used to break the list into individual sentences and assign one sentence per row.

An attempt to run SBERT using a distributed computing approach was made, but due to worker caching permission issues and after several alternative attempts, we were unable to load SBERT on each worker locally - the notebook constantly crashed. More specifically, the workers had restricted access to their default caches, preventing them from loading the SBERT model from Hugging Face. A temporary cache for each worker was attempted; however, Hugging Face's library inconsistently applied the override and occasionally still tried to load it into the worker's default cache, causing the code to break. Regardless, documentation of the intended distributed approach will be made to support our methodology.

The SBERT model used was all-MiniLM-L6-v2. To enable batch sentence processing, @pandas_udf was used, and the resulting batches were then input into the SBERT model using encode(). Taking a batch-processing approach allows the distributed computing process to run more efficiently by enabling SBERT to encode several sentences at once rather than ~1.2 million sentences one by one. Considering this, a different approach was implemented to run the model. Due to workers' restricted access, the embeddings were run solely using the master node. To address memory issues, data was partitioned into 200,000-sentence chunks and encoded into the SBERT model. As a defense against GCP crashing, embedded chunks were saved to the bucket at the end of each loop. We ended up with 6 saved embedded files, which we combined into a single complete array.

Cosine Similarity and Topic Diversity Index (NB02_feature_extraction)

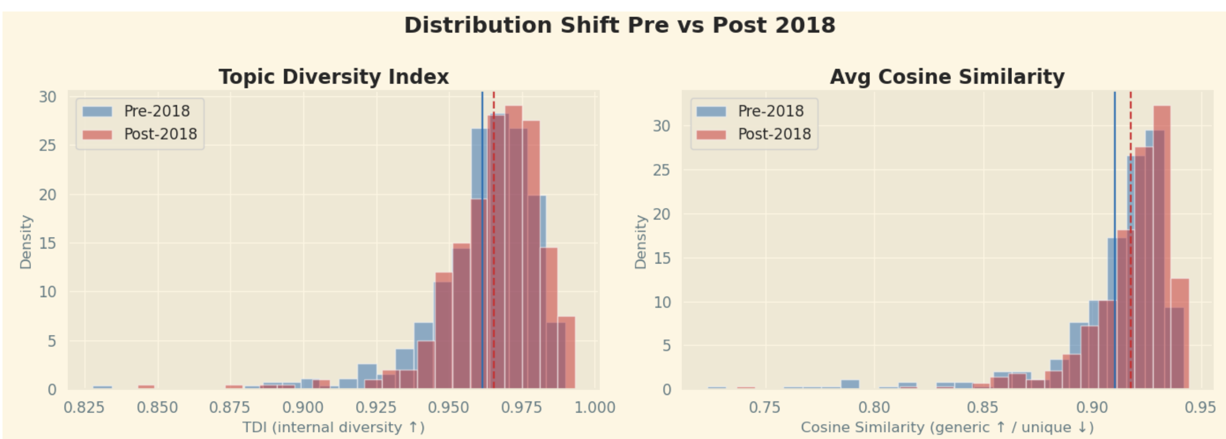
Using film-level subtitle features from sentence embeddings, we stored and loaded them in BigQuery because they exceed 1.7GB. It first filters out low-information subtitle lines, keeping sentences with at least 3 words and 10 characters while excluding noisy symbol-heavy or numeric-only text. This reduces the corpus from 1,121,360 subtitle rows to 795,119 usable rows (70.9%), then aggregates them into 746 films with an average embedding per film.

Two main metrics are created. Mean cosine similarity compares each film's average embedding against every other film's average embedding; higher values suggest a film is more semantically "generic" or like the rest of the corpus. Across 746 films, the mean is 0.9133, with a range from 0.7233 to 0.9443. The highest-scoring examples include Fast & Furious Presents: Hobbs & Shaw (0.9443), Thunderbolts (0.9419), and Avengers: Endgame (0.9419).

The second core metric is TDI, the Topic Diversity Index. It samples up to 500 sentences per film, clusters 364,254 sentence embeddings using MiniBatch K-Means, selects K=15 topics based on the highest silhouette score, and computes normalized Shannon entropy over each film's topic distribution. Higher TDI means dialogue is spread more evenly across topics; lower TDI means more topic concentration/repetition. TDI has mean 0.9634, min 0.8277, and max 0.9931. The most topic-diverse films by TDI include Zombieland: Double Tap (0.9931), Thunderbolts (0.9920), and Nope (0.9902), while the least topic-diverse include The Great Wall (0.8277) and Greyhound (0.8429).

NUMERICAL RESULTS

Exploratory Data Analysis (NB03_eda)

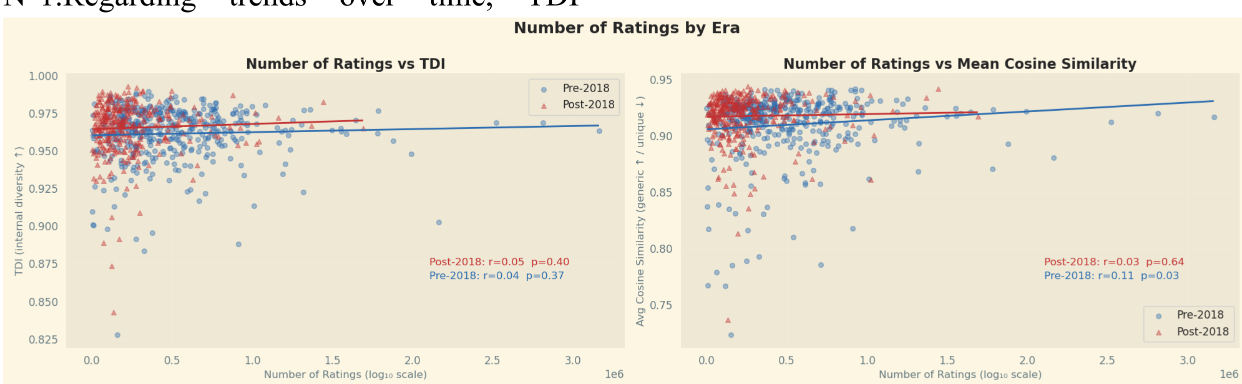


Plot N°1. Distribution Shift Pre vs Post 2018

In our corpus, the most topically repetitive films (lowest TDI) tend to be narratively focused and genre-driven — sci-fi, thrillers or crime — in which dialogue clusters tightly around a single dominant topic. In contrast, the most diverse films (highest TDI) are predominantly ensemble or franchise productions (Thunderbolts*, Ant-Man and the Wasp, Pitch Perfect) whose scripts span romance, action and humour simultaneously. Notably, topically diverse films also display higher mean cosine similarity, suggesting that despite their internal variety, they closely resemble other films in the corpus — consistent with the formulaic nature of many franchise entries.

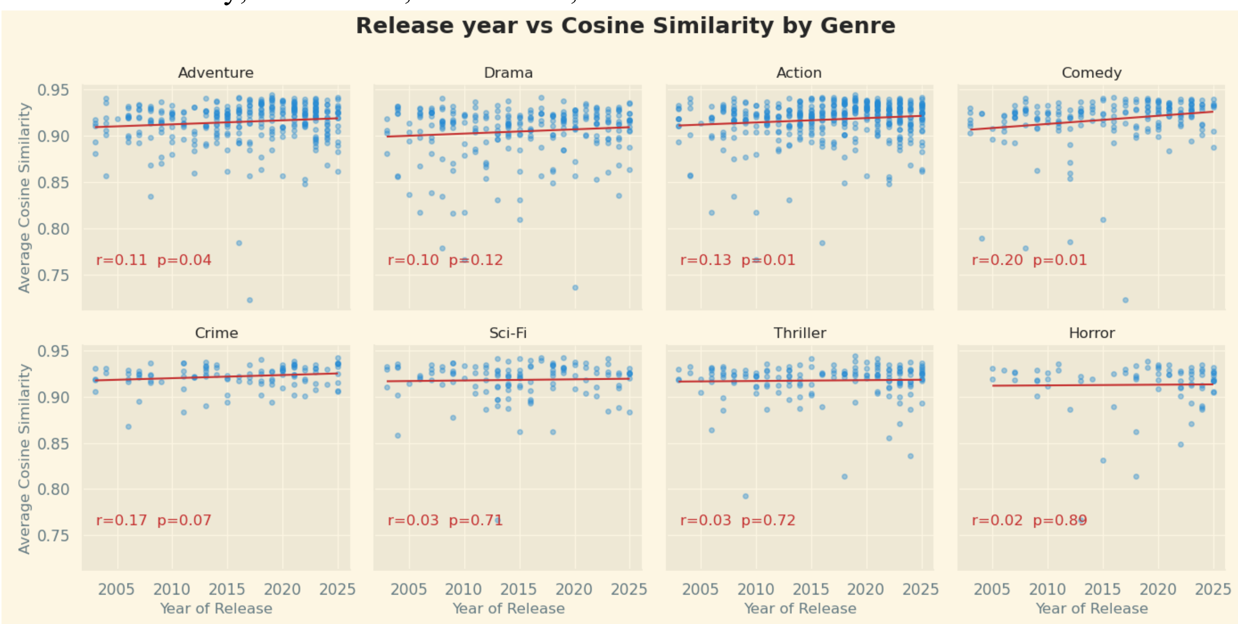
Turning to cosine similarity, the most unique films show no clear thematic pattern — spanning comedies (Ted), dramas (The Big Short), and children's films (Cars 3) — suggesting that low similarity reflects stylistic or linguistic consistency rather than genre. The most generic films (highest cosine similarity), by contrast, are almost exclusively large-scale franchise productions (Avengers: Endgame, Logan, Iron Man 3) whose scripts blend action, humour, emotion, and exposition in ways that diverge significantly from the corpus mean. These films also carry the highest vote counts and box office figures. Both distributions show a modest rightward shift post-2018, indicating that films have become marginally more topically diverse and linguistically similar to the corpus over time, though the shift remains subtle rather than dramatic, as shown in Plot N°1.Regarding trends over time, TDI remains highly stable at 0.960-0.965 throughout the pre-2018 period, with only a minor increase after 2020. Mean cosine similarity, by contrast, shows a more dynamic pattern, rising steadily from 2013 to 2020 before stabilizing in a higher range from 2021 onwards. Time is positively related to higher TDI and Cosine Similarity, meaning that movies have more dialogue spanning a wide range of topics, but the meaning of the dialogue has also become more generic.

Neither TDI nor cosine similarity shows a meaningful correlation with runtime, suggesting that film length does not drive topical diversity or linguistic uniqueness. Similarly, TDI shows no relationship with audience engagement, as measured by the logarithm of the number of ratings. Cosine similarity, however, tells a different story: among pre-2018 films, higher vote counts are associated with greater linguistic similarity, suggesting that more widely seen films tended to conform to a shared cinematic language as shown in Graphic. This relationship disappears entirely in post-2018 films, where no such pattern is observed, potentially reflecting a fragmentation of mainstream cinema in which popular films no longer converge on a common linguistic style.



Plot N° 2. Number of Ratings by Era

Furthermore, when analyzing the mean cosine similarity by genre, we observe that approximately half of the genres show no significant variation between pre- and post-2018 movies. However, some genres increased their cosine similarity after 2018, such as fantasy, adventure, animation, mystery, action, and biography, as seen in Plot N°3. Across genres, TDI shows relatively little variation between the pre- and post-2018 periods, though drama and fantasy exhibit a notable increase in topical diversity after 2018.



Plot N°3. Release Year vs Cosine Similarity by Genre

Regression analyses (NB04_regression)

The linear regression analyses show that linguistic genericness and topical diversity in film scripts are shaped by distinct forces. For Mean Cosine Similarity, scripts with more dialogue lines are markedly more generic, indicating that longer scripts converge on a shared cinematic vocabulary. Holding dialogue count constant, longer runtimes make films less generic, suggesting that runtime and dialogue density capture through a different channel. Language has also become more generic over time, even after accounting for genre, age certification, and popularity. Crime, Horror, and Romance scripts are the most linguistically generic, whereas Drama stands out as more idiosyncratic, and higher age certifications—especially PG—are associated with more generic language, consistent with broad-audience optimisation.

For the Topic Diversity Index, the temporal trend toward genericness persists, but runtime no longer matters, implying that topical diversity depends more on content than duration. Genre effects partially realign:

Horror and Romance remain highly diverse, Comedy and Crime gain prominence, and Fantasy and Adventure—previously strong for linguistic similarity—no longer predict topical diversity, implying these genres are linguistically generic but thematically varied. The age-certification pattern largely disappears for TDI, with only PG remaining significant and R-rated films showing virtually no effect, indicating that rating shapes how films speak far more than what they speak about. In both regressions, the year coefficient is positive and significant, confirming that the gradual drift toward linguistic genericness and topical diversity has increased over time, even after controlling for genre, certification, and popularity.

CONCLUSIONS

The findings do not support the original hypothesis that post-2018 films became more narratively repetitive or thematically concentrated in response to the rise of short-form video platforms. If the attention-span hypothesis were strongly reflected in subtitle text, we would expect lower topic diversity and greater semantic repetition after 2018. Instead, the results show a modest increase in Topic Diversity Index over time, suggesting that recent films tend to distribute dialogue across a slightly broader range of topics rather than fewer ones.

At the same time, mean cosine similarity increases over time, indicating that films have become more semantically similar to the wider corpus. This suggests a more nuanced pattern: contemporary films may be becoming linguistically more generic while also covering a wider set of internal topics. In other words, films may increasingly rely on shared cinematic language or formulaic dialogue patterns, even when their scripts span multiple themes, genres, or narrative functions.

Overall, the study finds limited evidence for a TikTok-driven decline in narrative complexity. Instead, it identifies a gradual movement toward films that are simultaneously more linguistically conventional and slightly more topically diverse. Future research should test this relationship using full screenplays rather than subtitles, include stronger controls for genre composition, franchise status, budget, and streaming-era distribution, and apply a formal interrupted time-series design before making causal claims about short-form video's influence on cinema.

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Team Collaboration

- This project was carried out through equal and sustained contribution from all four team members across every stage of the research pipeline. Responsibilities were distributed collaboratively rather than partitioned in isolation, with each member actively participating in data collection, preprocessing, modelling decisions, and the interpretation of results. Regular communication was maintained throughout the project via group meetings and shared documentation, ensuring that methodological decisions were discussed and agreed upon collectively.

Gen AI

- Generative AI tools, specifically Claude by Anthropic and Copilot, were used throughout this project as a technical assistant to support code development, error resolution, and metric design. Given the computational constraints of working on Google Cloud Platform — including repeated kernel crashes caused by attempting to load millions of high-dimensional embeddings into memory, authentication issues with distributed SBERT inference across worker nodes, and BigQuery syntax errors. These tools were good in diagnosing failure modes and proposing memory-efficient alternatives, such as pushing mean-pooling aggregation into BigQuery SQL to avoid materialising large matrices locally. All AI-generated suggestions were critically evaluated, adapted, and validated by the team before incorporation into the final pipeline.
  - Prompts used by Carlos Mora:
    - https://claude.ai/share/a04f1b54-9504-42cf-9e80-a3483e5abe05